



Lecture 3



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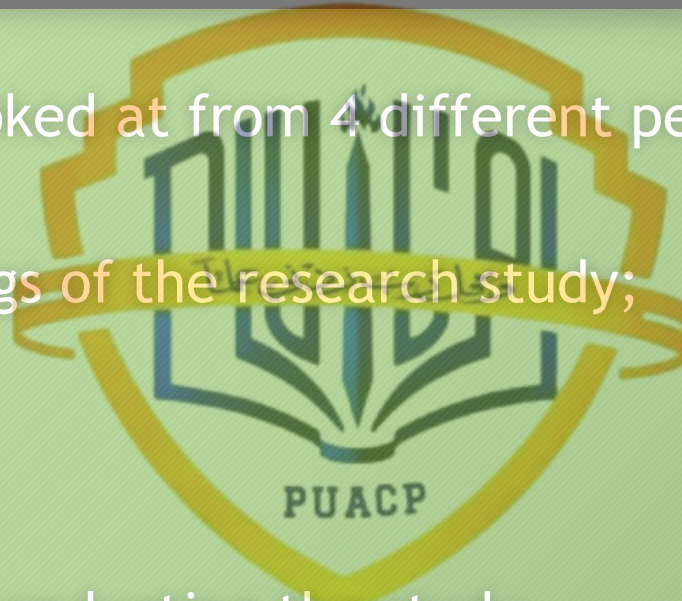
Types of Research



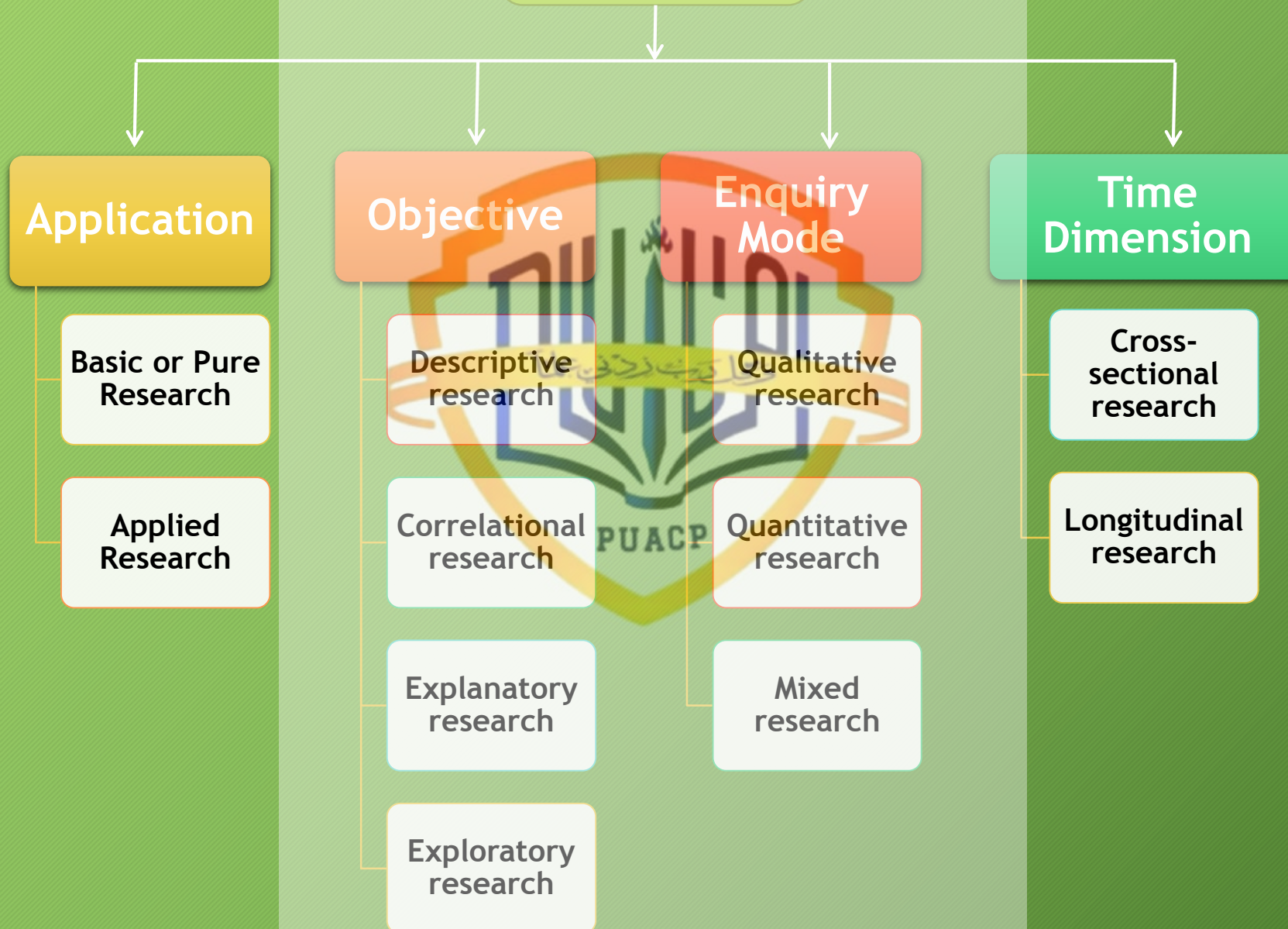
Types of research

Types of research can be looked at from 4 different perspectives

1. Applications of the findings of the research study;
2. Objectives of the study;
3. Mode of enquiry used in conducting the study
4. Time dimension

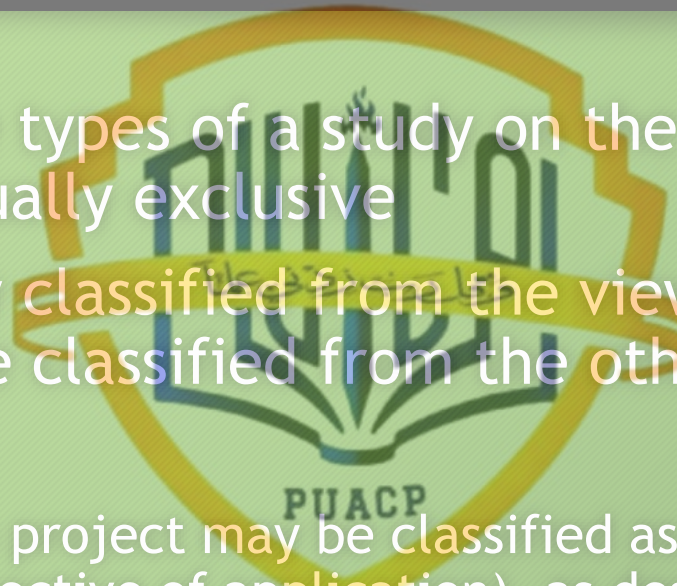


Classification of Research



Point to Remember

- The classification of the types of a study on the basis of these perspectives is not mutually exclusive
- that is, a research study classified from the viewpoint of 'application' can also be classified from the other perspectives.
- **For example**, a research project may be classified as pure or applied research (from the perspective of application), as descriptive, correlational, explanatory or exploratory (from the perspective of objectives) and as qualitative or quantitative (from the perspective of the enquiry mode employed).



Types of research: application perspective



Application perspective

From the perspective of application there are two broad categories:

- Pure research &
- Applied research

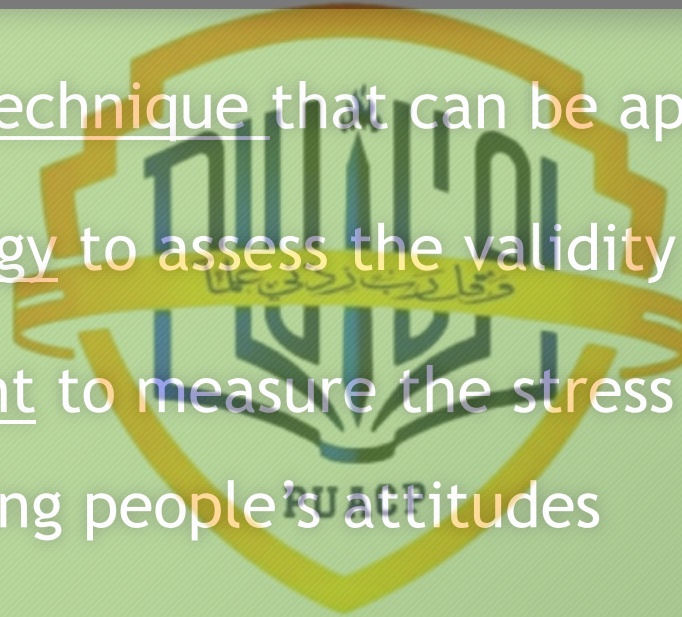


Pure research

- It is an investigation on basic principles and reasons for occurrence of a particular event or process
- It involves developing, improving and testing theories, methods, techniques and tools that form the body of research methodology
- It may or may not have practical application at the present time or in the future.
- It helps build new frontiers of knowledge and add something new to the existing body of knowledge of research methods.

Examples

- Developing a sampling technique that can be applied to a particular situation
- developing a methodology to assess the validity of a procedure
- developing an instrument to measure the stress level in people and finding the best way of measuring people's attitudes



Applied research

- Most of the research in the social sciences is applied.
- A research, the outcome of which has immediate application is also termed as applied research.
- The research techniques, procedures and methods that form the body of research methodology are applied.
- In an applied research one solves certain problems employing well known and accepted theories and principles.

Difference b/w Basic and Applied Research

Basic research	Applied research
Seeks generalization	Studies individual or specific cases without the objective to generalize
Aims at basic processes	Aims at any variable which makes the desired difference
Attempts to explain why things happen	Tries to say how things can be changed
Tries to get all the facts	Tries to correct the facts which are problematic
Reports in technical language of the topic	Reports in common language

Types of research: Objectives perspective



Objectives perspective

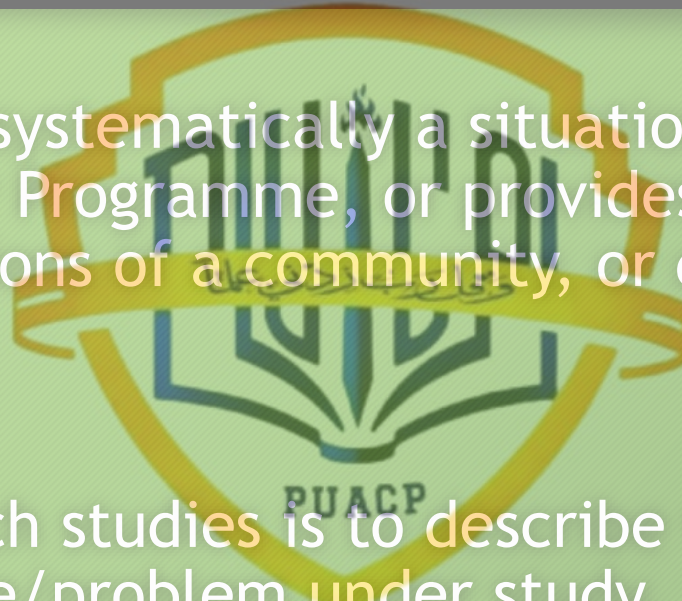
From the perspective of objectives, broadly a research can be classified as

- Descriptive,
- Correlational,
- Explanatory or
- Exploratory.



Descriptive Study

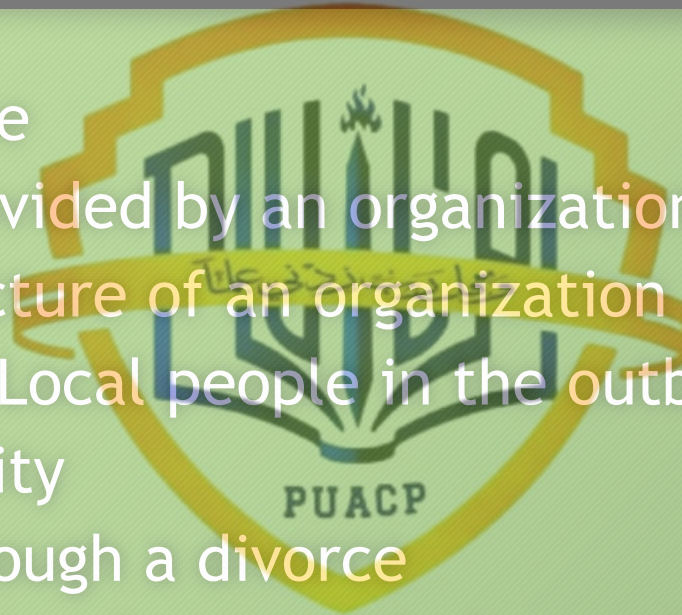
- It attempts to describe systematically a situation, problem, phenomenon, service or Programme, or provides information about, the living conditions of a community, or describes attitudes towards an issue.
- The main purpose of such studies is to describe what is prevalent with respect to the issue/problem under study.



Examples

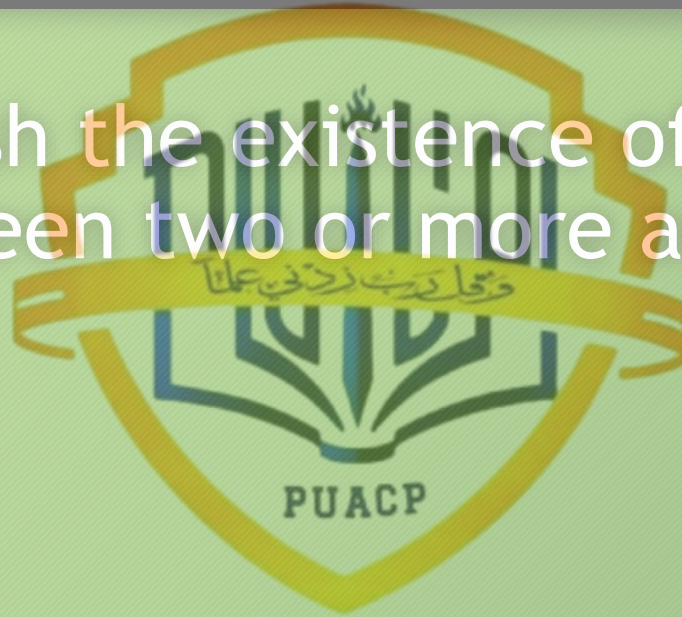
It may attempt to describe

- The types of service provided by an organization
- The administrative structure of an organization
- The living conditions of Local people in the outback
- The needs of a community
- What it means to go through a divorce
- How a child feels living in a house with domestic violence
- The attitudes of employees towards management.



Correlational study

- A study to establish the existence of a relationship between two or more aspects of a situation.

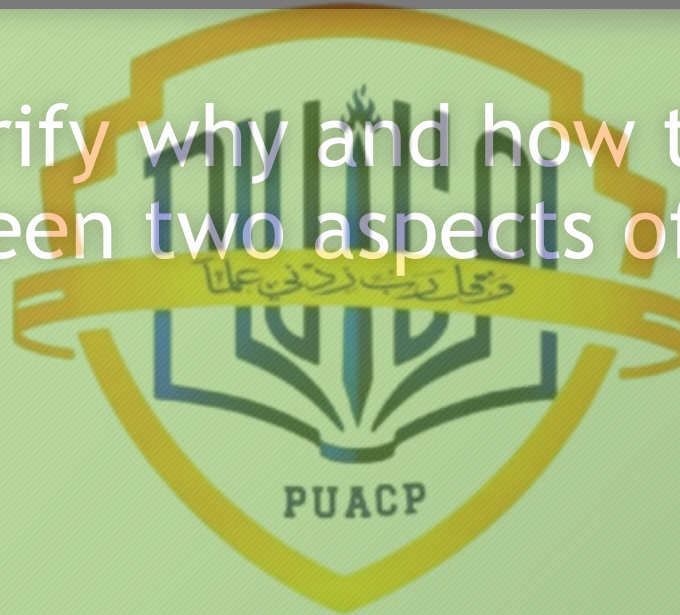


Examples

- What is the impact of an advertising campaign on the sale of a product?
- What is the relationship between stressful living and the incidence of heart attack?
- What is the relationship between fertility and mortality?
- What is the relationship between technology and unemployment?
- What is the effect of a health service on the control of a disease?
- What is the effect of the home environment on educational achievement?

Explanatory research

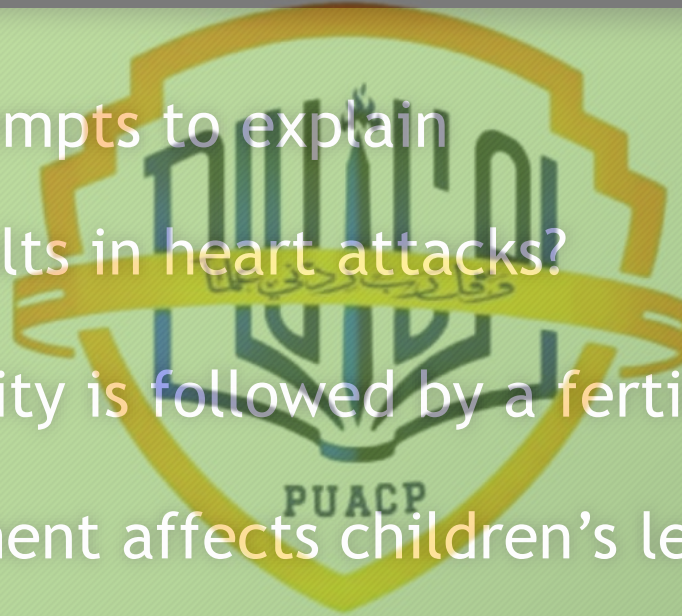
- It attempts to clarify why and how there is a relationship between two aspects of a situation or phenomenon.



Examples

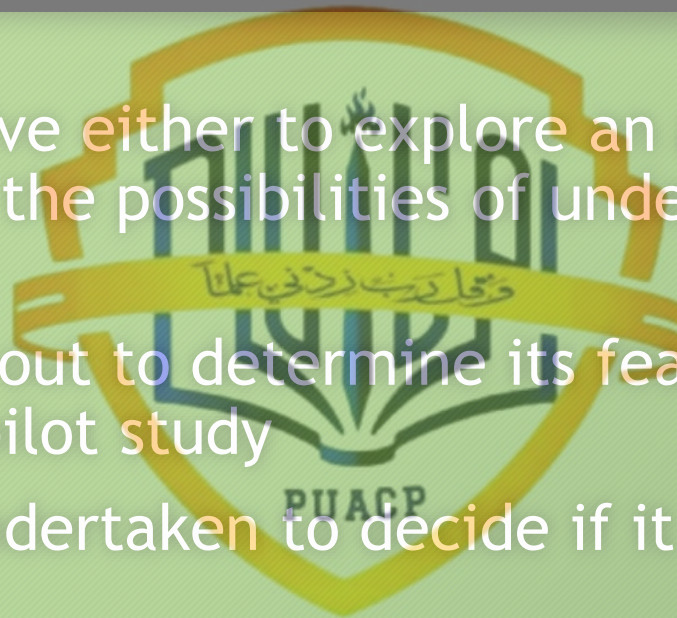
This type of research attempts to explain

- Why stressful living results in heart attacks?
- Why a decline in mortality is followed by a fertility decline?
- How the home environment affects children's level of academic achievement?



Exploratory research

- A study with the objective either to explore an area where little is known or to investigate the possibilities of undertaking a particular research study
- When a study is carried out to determine its feasibility it is also called a feasibility study or a pilot study
- A small-scale study is undertaken to decide if it is worth carrying out a detailed investigation.
- On the basis of the assessment made during the exploratory study, a full study may eventuate.
- Exploratory studies are also conducted to develop, refine and/or test measurement tools and procedures.



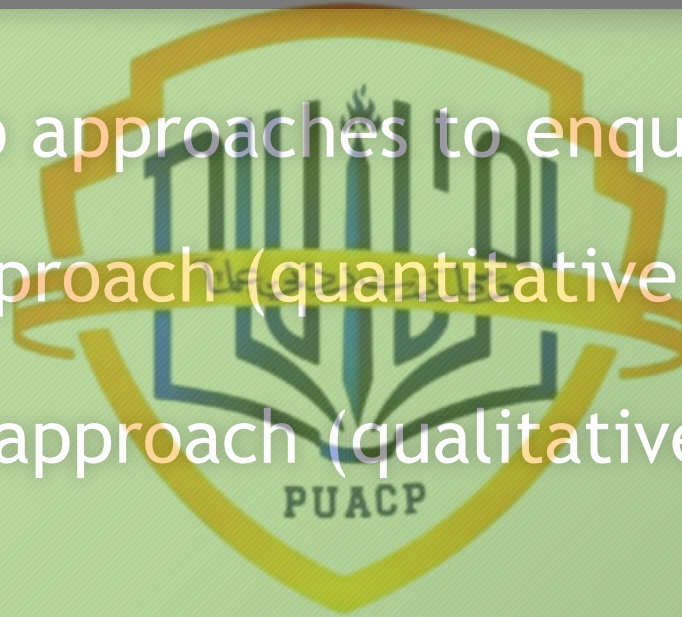
Types of research: Enquiry perspective



Mode of enquiry perspective

Broadly, there are two approaches to enquiry

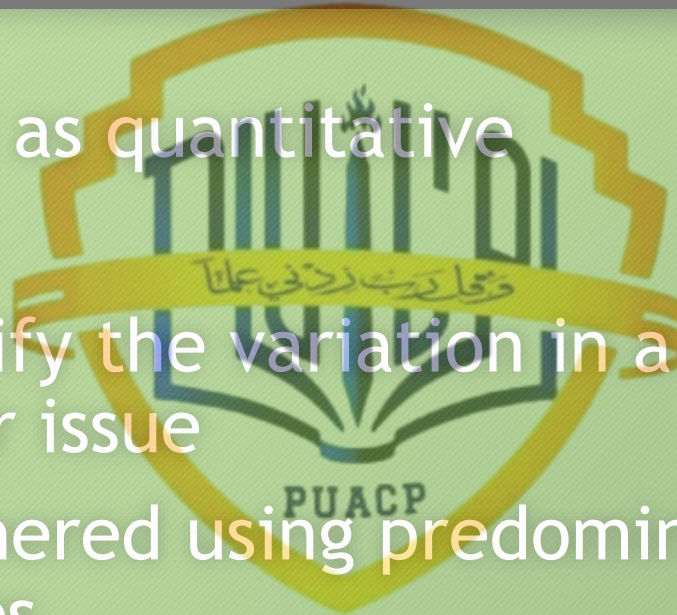
- 1. The structured approach (quantitative research)
- 2. The unstructured approach (qualitative research)



quantitative research/ Structured approach

The study is classified as quantitative

- if you want to quantify the variation in a phenomenon, situation, problem or issue
- if information is gathered using predominantly quantitative variables
- if the analysis is geared to ascertain the magnitude of the variation



Examples of quantitative research

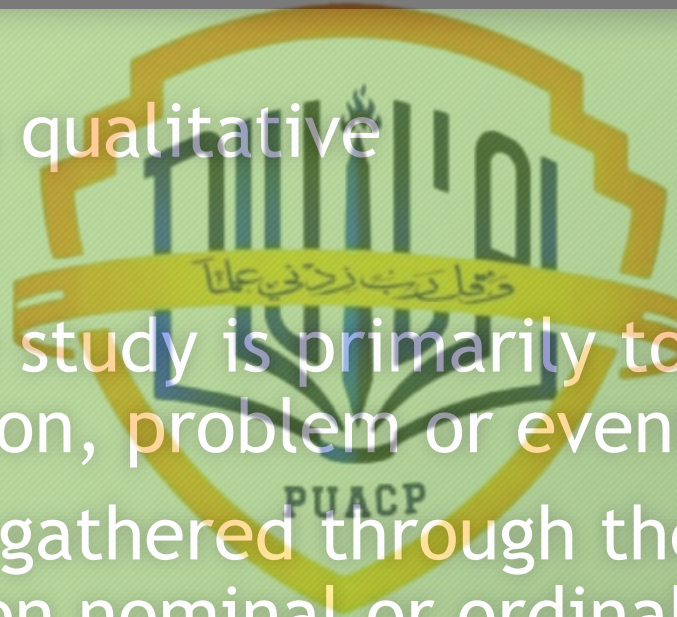
- How many people have a particular problem?
- How many people hold a particular attitude?



qualitative research/ unstructured approach

A study is classified as qualitative

- if the purpose of the study is primarily to describe a situation, phenomenon, problem or event
- if the information is gathered through the use of variables measured on nominal or ordinal scales (qualitative measurement scales)
- if the analysis is done to establish the variation in the situation, phenomenon or problem without quantifying it



Examples of qualitative research

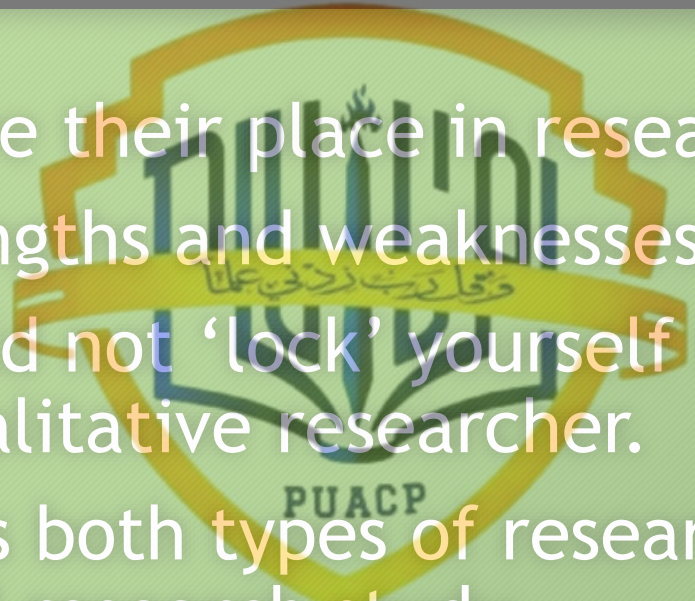
- The description of an observed situation
- The historical enumeration of events
- An account of the different opinions people have about an issue
- A description of the living conditions of a community



Quantitative research	Qualitative research
Purpose is to quantify the extent of variation in a situation, issue	Purpose is to describe variation in a situation, issue
Emphasis on some form of measurement of variable	Emphasis on description of variables
Emphasis on greater sample size	Fewer cases
Narrow focus in terms of extent of inquiry from large number of respondents	It covers multiple issues from fewer respondents
Analysis of data through frequency distribution cross tabulations and other statistical procedures	Analysis involves narratives, identification of themes
Disciplines like psychology, epidemiology, education, economics, public health and marketing are more inclined towards quantitative research.	Disciplines like anthropology, history and sociology are more inclined towards qualitative research

Important to Note

- Both approaches have their place in research.
- Both have their strengths and weaknesses.
- Therefore, you should not 'lock' yourself solely into a quantitative or a qualitative researcher.
- In the social sciences both types of research are important for a good research study



Furthermore

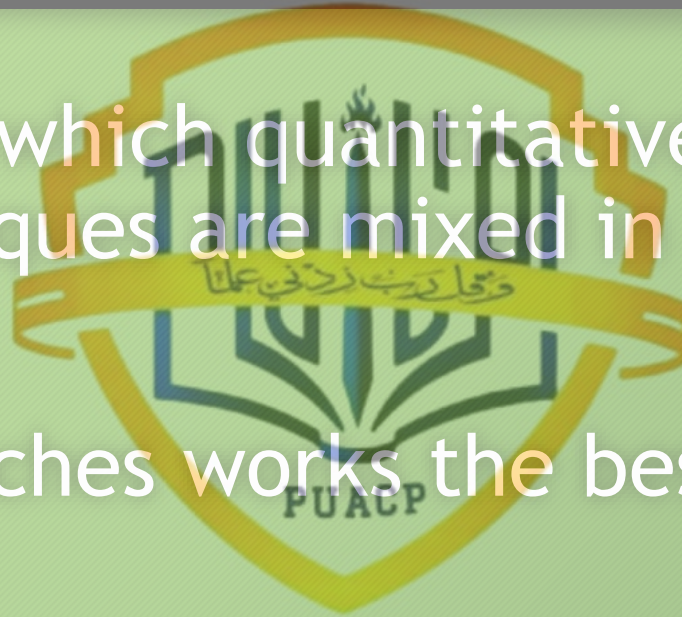
The choice between quantitative and qualitative approaches should depend upon:

- Aim of your enquiry - exploration, confirmation or quantification.
- Use of the findings - policy formulation or process understanding.



Mixed Research Method

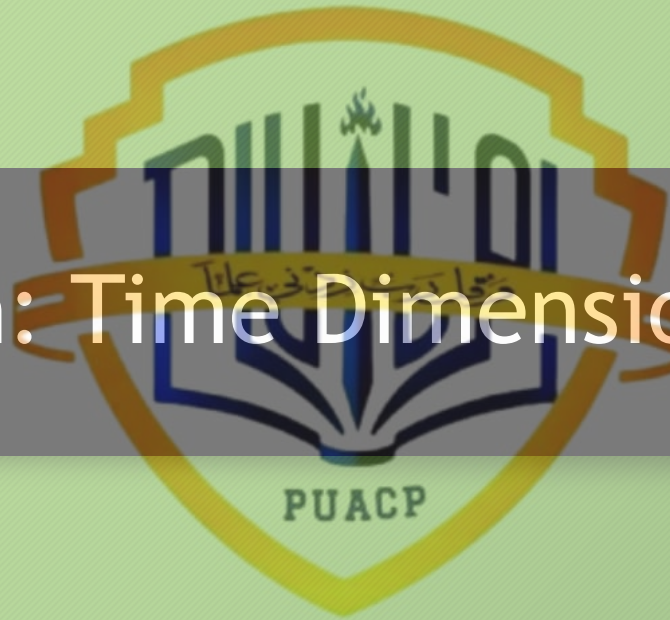
- It is a research in which quantitative and qualitative techniques are mixed in a single study.
- Mixture of approaches works the best in real world situation.



Example

- suppose you want to find out the types of service available to victims of domestic violence in a city and the extent of their utilization.
- Types of service is the qualitative aspect of the study as finding out about them entails description of the services.
- The extent of utilization of the services is the quantitative aspect as it involves estimating the number of people who use the services and calculating other indicators that reflect the extent of utilization.

Types of research: Time Dimension



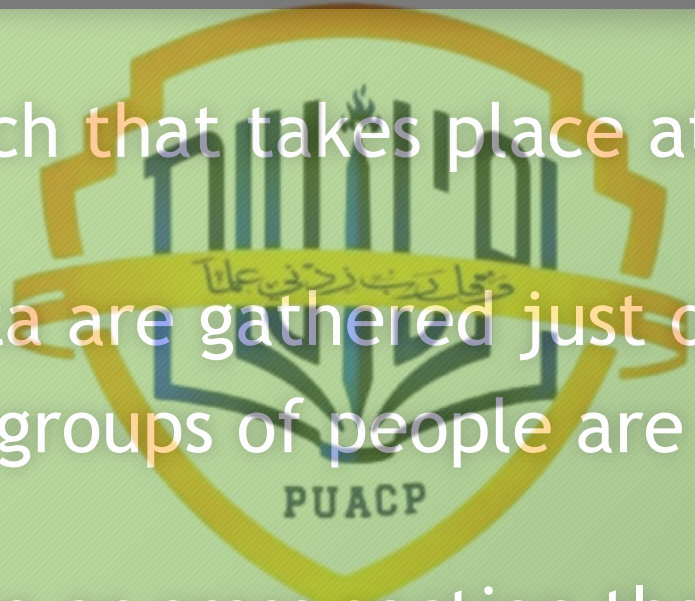
From the perspective of time dimension there are two broad categories:

- Cross sectional research and
- Longitudinal research



Cross sectional research

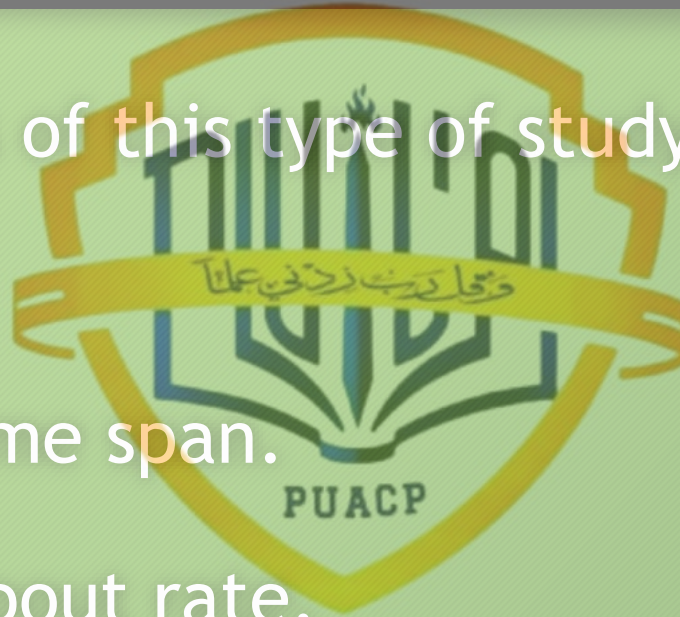
- It is a type of research that takes place at a single point in time.
- In such study the data are gathered just once.
- In this study several groups of people are examined at one point in time.
- It is like taking a slice or cross section that we are measuring or observing.



Advantages of cross sectional research

The major advantages of this type of study are:

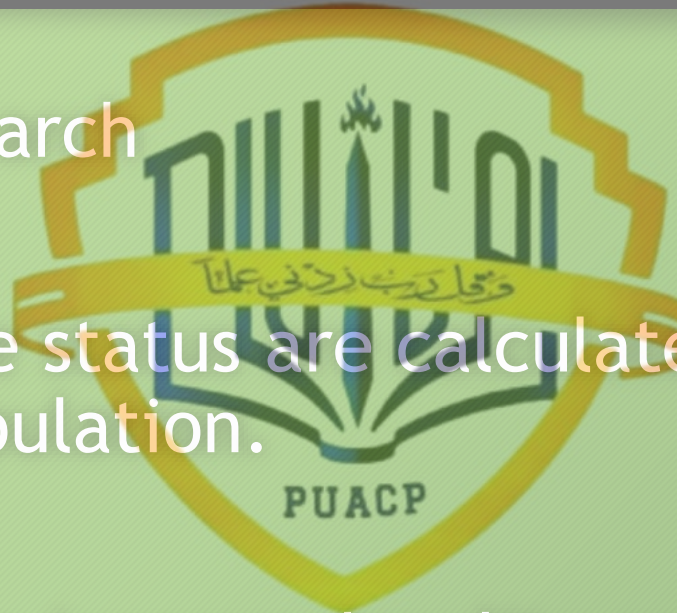
- It is inexpensive.
- It involves shorter time span.
- There is a lower dropout rate.



Examples

In cross sectional research

- Disease and exposure status are calculated at the same time in a certain population.
- Relationship between income, locality and expenditure at point in time are being studied.

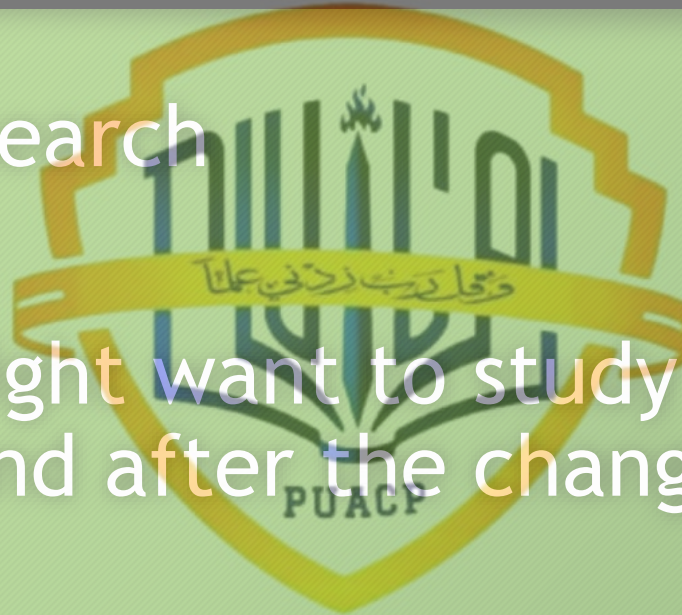


Longitudinal Research

- It is a type of study that takes place over time.
- In such studies researcher want to study people or phenomena at more than one point in time.
- When the data on dependent variable are gathered at two or more points in time then it is called longitudinal research.
- It acquires more time and cost than cross sectional study.

Examples

- In longitudinal research
- The researcher might want to study employees behavior before and after the change in top management.



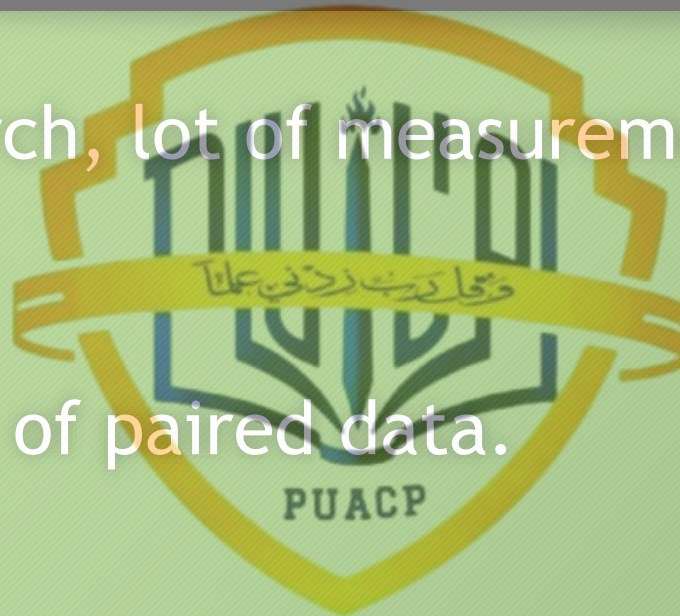
Types of Longitudinal research

- The longitudinal study designs are further divided into two major types
- Repeated Measures
- Time Series



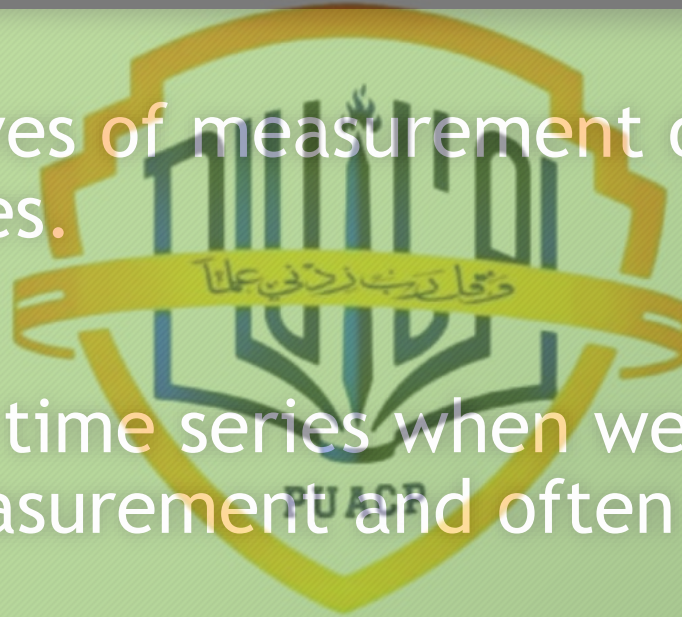
Repeated Measure Design

- In this type of research, lot of measurements are made on same individual.
- It is a generalization of paired data.



Time Series

- If we have many waves of measurement over time then it is called time series.
- Usually we use term time series when we have at least twenty waves of measurement and often far more.



Further types of longitudinal studies

- **Cohort Study:** it defines as a group experiencing some events in a selected time period and studying them at intervals through time.
- **Panel Study:** Sample a cross-section, and survey it at regular intervals.
- **Retrospective Study:** A study that looks back in time for example searching for medical records or preceding years to try to find a trend.
- **Prospective Study:** A longitudinal study in which data are collected at two or more time points moving forward on one or more variables.

Thanks

